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Correction to "The Conditional CRO Trilemma: Impossibility Bounds on Confidentiality, Reliability, and Opposability Under Institutional Extraction Limits"

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In the above article [1], the statement of Lemma 4 (Confidentiality–Mutual Information Trade-off) contains an error in the direction of the inequality, and Steps 3–6 of the proof of Theorem 1 rely on this statement and on an optimization step that is incorrectly solved. These errors were identified by the authors during the machine-checked formalization of the results in Lean 4. This correction (i) restates Lemma 4 in the form actually established by its proof; (ii) makes explicit the verdict-efficiency condition used in Step 3; (iii) states explicitly the leakage regime in which Theorem 1 holds and gives a complete proof with explicit constants; and (iv) corrects Remark 10 and clarifies the reading of Section VI-B. The institutional model (Assumptions 1–3), the role of the institutional parameters (Δ, α, β), and the form of the main bound $\mathsf{Conf} \cdot \mathsf{Rel} \cdot \mathsf{Opp} \le \Delta/\lambda + O(\lambda^{-2}) + \mathsf{negl}(\lambda)$ are preserved within the stated regime. The authors apologize for any inconvenience caused.

1 Correction of Lemma 4 (Section V-A)

The proof of Lemma 4 combines Pinsker's inequality with β-regularity (Assumption 3) and yields, for every pair (m_0, m_1) and context c,

$$\mathrm{TV}^2\big(P_{\Sigma|m_0,c}, P_{\Sigma|m_1,c}\big) \le c_0\, \mathrm{D}_{\mathrm{KL}}\big(P_{\Sigma|m_0,c} \,\|\, P_{\Sigma|m_1,c}\big) \le c_0\, \beta\, I(M;\Sigma \mid C),$$

that is, an *upper* bound on the distinguishing advantage, hence a *lower* bound on Conf (as written in (22)). The published statement (17) asserts an upper bound on Conf, which does not follow from this argument and does not hold in general; in addition, the normalization by λ in (21) is not justified. Lemma 4 and its proof are therefore replaced by the following.

Lemma 4 (Confidentiality guarantee for regular protocols)**.** *For any β-regular protocol $\mathcal{P}$ (Assumption 3),*

$$\mathsf{Conf}(\mathcal{P}) \;\ge\; 1 - \sqrt{c_0\, \beta\, I(M;\Sigma \mid C)}\,. \tag{17$'$}$$

Proof. By Lemma 1 and Assumption 3, for all $m_0, m_1 \in \mathcal{M}_\lambda$ and $c \in \mathcal{C}_\lambda$, $\mathrm{TV}(P_{\Sigma|m_0,c}, P_{\Sigma|m_1,c}) \le \sqrt{c_0\, \mathrm{D}_{\mathrm{KL}}(P_{\Sigma|m_0,c} \| P_{\Sigma|m_1,c})} \le \sqrt{c_0\, \beta\, I(M;\Sigma \mid C)}$. Taking the maximum over (m_0, m_1, c) and applying Definition 7 gives (17$'$). □

Remark (Replaces Remark 7). Lemma 4 states that low conditional mutual information, together with β-regularity, *guarantees* confidentiality: if $I(M;\Sigma \mid C) \le \kappa/\lambda$, then $\mathsf{Conf}(\mathcal{P}) \ge 1 - \sqrt{c_0 \beta \kappa/\lambda} \to 1$ as $\lambda \to \infty$.

Remark (Constant c_0, Remark 1). With divergences in bits, the tight Pinsker constant is $(\ln 2)/2 \approx 0.347$. The constant $c_0 = 1/(2 \ln 2) \approx 0.721$ used throughout the article is larger, so every inequality stated with c_0 remains valid; it is, however, not tight, and the word "tight" in Remark 1 should be deleted.

2 Verdict efficiency (Step 3 of the proof of Theorem 1)

Inequality (28) states $H(V \mid C) \ge \log_2 k_\lambda - 1 \ge \alpha H(C) - 1$. Assumption 2 gives $\log_2 k_\lambda \le \alpha H(C)$, an *upper* bound, so the second inequality in (28) does not follow from Assumption 2. The lower bound on $H(V \mid C)$ is the efficiency property of verdict functions discussed in Section IV-B.2 ("verdicts effectively utilize the available context"), and it is now stated explicitly as Assumption 2$'$.

Assumption 2′ (Bounded and efficient verdict space; replaces Assumption 2)**.** *The institutional verdict space satisfies $k_\lambda \le 2^{\alpha H(C)}$ for some constant α, and the verdict function uses the available context efficiently:*

$$H(V \mid C) \;\ge\; \alpha H(C) - 1 \;=:\; \gamma, \qquad \gamma \ge 1. \tag{11′}$$

For the institutional settings considered in Section IV-B ($H(C) \ge 10$ bits, $\alpha \approx 1$), one has $\gamma \ge 9$.

3 Corrected statement and proof of Theorem 1 (Section V-B)

Remark 10 identifies the regime $I(M;\Sigma \mid C) = O(1/\lambda)$ as the relevant operating regime of the trilemma. In the published proof this regime was obtained from the optimization (33)–(34); that computation is incorrect ($f'(x) = 2x - \frac{5}{2}ax^{3/2}$ vanishes at $\sqrt{x^*} = 4/(5a)$, not $5a/4$, and x^* is a maximizer of f). The regime is therefore stated explicitly as an assumption. By Lemma 4 it is precisely the regime in which regular protocols guarantee confidentiality.

Assumption 4 (Confidentiality-preserving regime)**.** *There exists a constant $\kappa \ge 0$, independent of λ, such that*

$$I(M;\Sigma \mid C) \;\le\; \frac{\kappa}{\lambda}. \tag{12′}$$

Theorem 1 (Conditional CRO Trilemma, corrected)**.** *Under Assumption 1 (bounded extraction), Assumption 2′ (11′), Assumption 3 (β-regularity) and Assumption 4 (12′), for any evidence protocol $\mathcal{P}$ with security parameter $\lambda \ge 64$, message entropy $H(M) = \lambda$, M independent of C, and institutional parameters (Δ, α):*

$$\mathsf{Conf}(\mathcal{P}) \cdot \mathsf{Rel}(\mathcal{P}) \cdot \mathsf{Opp}(\mathcal{P}) \;\le\; \frac{1}{\gamma}\left(\frac{\Delta}{\lambda} + \frac{\kappa(\Delta+1)}{\lambda^2} + \frac{\kappa^2}{\lambda^3}\right) \;\le\; \frac{\Delta}{\lambda} + O(\lambda^{-2}). \tag{23′}$$

In particular, the published bound $\mathsf{Conf} \cdot \mathsf{Rel} \cdot \mathsf{Opp} \le \Delta/\lambda + O(\lambda^{-2}) + \mathsf{negl}(\lambda)$ holds under these assumptions. Moreover, $\mathsf{Conf}(\mathcal{P}) \ge 1 - \sqrt{c_0 \beta \kappa/\lambda}$ by Lemma 4.

Proof. Write $x = I(M;\Sigma \mid C)$.

Step 1 (reliability forces mutual information). By Fano's inequality (Lemma 2) applied to the optimal estimator $\hat{M}$ of M from (Σ, C), with $p_e = 1 - \mathsf{Rel}$ and $h(p_e) \le 1$,

$$H(M \mid \Sigma, C) \le 1 + p_e \log_2(2^\lambda - 1) \le 1 + (1 - \mathsf{Rel})\,\lambda.$$

Hence $I(M;\Sigma,C) \ge \mathsf{Rel} \cdot \lambda - 1$, and since $M \perp C$, $x = I(M;\Sigma,C)$. Therefore

$$\mathsf{Rel}(\mathcal{P}) \le \frac{x+1}{\lambda}. \tag{26′}$$

This holds for every value of Rel; the restriction $\mathsf{Rel} \ge 1/2$ used before (25) is not needed.

Step 2 (opposability through institutional extraction). By Definition 9, Assumption 1 and (11′),

$$\mathsf{Opp}(\mathcal{P}) = \frac{I(V;\Sigma \mid C)}{H(V \mid C)} \le \frac{x+\Delta}{\gamma}. \tag{29′}$$

Step 3 (product bound). Since $0 \le \mathsf{Conf}(\mathcal{P}) \le 1$ and all factors are nonnegative,

$$\mathsf{Conf} \cdot \mathsf{Rel} \cdot \mathsf{Opp} \;\le\; \frac{x+1}{\lambda} \cdot \frac{x+\Delta}{\gamma}.$$

The right-hand side is nondecreasing in x; by (12′),

$$\mathsf{Conf} \cdot \mathsf{Rel} \cdot \mathsf{Opp} \;\le\; \frac{1}{\gamma}\Big(\frac{1}{\lambda} + \frac{\kappa}{\lambda^2}\Big)\Big(\Delta + \frac{\kappa}{\lambda}\Big) = \frac{1}{\gamma}\Big(\frac{\Delta}{\lambda} + \frac{\kappa(\Delta+1)}{\lambda^2} + \frac{\kappa^2}{\lambda^3}\Big),$$

which is (23′); the last inequality in (23′) uses $\gamma \ge 1$. The lower bound on Conf is Lemma 4 combined with (12′). □

Equations (30)–(39) and Remark 10 of the original article are replaced by the above proof. Corollary 1 ($\mathsf{Conf} \cdot \mathsf{Rel} \cdot \mathsf{Opp} \lesssim \Delta/\lambda$ for $\lambda \ge 128$, $\beta = O(1)$, $\alpha = O(1)$) holds under Assumptions 1, 2′, 3 and 4. Remark 11 (institutional capacity Δ as the dominant term) holds under the same assumptions.

Remark (Scope). Assumption 4 is essential. Without a bound on $I(M;\Sigma \mid C)$, the product $\mathsf{Conf} \cdot \mathsf{Rel} \cdot \mathsf{Opp}$ need not decay with λ: the protocol that reveals M with probability $1/2$ and otherwise outputs a uniformly random proof is β-regular with $\beta \le 1.04$ for all $\lambda \ge 64$, satisfies Assumptions 1–3, and has $\mathsf{Conf} = 1/2$, $\mathsf{Rel} = 1/2 + 2^{-\lambda-1}$ and, with V the first bit of Σ, $\mathsf{Opp} = 1$, for every λ. Here $I(M;\Sigma \mid C) = \lambda/2 - 1 + O(\lambda 2^{-\lambda})$ lies outside the regime (12′).

4 Clarification of Section VI-B

Table 2 was obtained by inverting Corollary 1 on the empirical products reported by the CLO framework [2]. These products are computed from CLO's operational scores, which are measured on a scale different from the information-theoretic quantities of Definitions 7–9. Under Assumption 4, Step 1 gives $\mathsf{Rel}(\mathcal{P}) \leq (1+\kappa/\lambda)/\lambda$ in the sense of Definition 8, so the empirical CLO products are not instances of $(23')$. The implied capacities of Table 2 and Remark 13 should therefore be read as an empirical calibration of Δ on the CLO scale, consistent with the bounded-rationality range of Section IV-B, and not as consequences of Theorem 1. The regime analysis of Section VI-A and the design recommendations of Section VI-C are unaffected.

References

[1] T. E. Minka Mi Nguidjoï, F. S. Mani Onana, and T. Djotio Ndié, "The conditional CRO trilemma: Impossibility bounds on confidentiality, reliability, and opposability under institutional extraction limits," *IEEE Access*, vol. 14, pp. 33263–33274, 2026, doi: 10.1109/ACCESS.2026.3669065.

[2] Reference [23] of [1] (Cryptographic Legal Opposability framework).